Lightweight Security-Aware Protocol Design for Resource-Constrained IoT

Sambu Srinidhi[a], [a,*] and Author-3[c]

[a]School of Computer Science and Engineering, VIT-AP University, Amaravathi. Andhra Pradesh, India

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ABSTRACT

The growing proliferation of devices on the Internet of Things (IoT) has made modern networks increasingly vulnerable to sophisticated cyber threats. Traditional intrusion detection systems (IDS) struggle to maintain high detection accuracy in the presence of high-dimensional data and extreme class imbalance, particularly for rare but critical attacks. The growing proliferation of devices on the Internet of Things (IoT) has made modern networks increasingly vulnerable to sophisticated cyber threats. Traditional intrusion detection systems (IDS) struggle to maintain high detection accuracy in the presence of high-dimensional data and extreme class imbalance, particularly for rare but critical attacks. The growing proliferation of devices on the Internet of Things (IoT) has made modern networks increasingly vulnerable to sophisticated cyber threats. Traditional intrusion detection systems (IDS) struggle to maintain high detection accuracy in the presence of high-dimensional data and extreme class imbalance, particularly for rare but critical attacks. The growing proliferation of devices on the Internet of Things (IoT) has made modern networks increasingly vulnerable to sophisticated cyber threats. Traditional intrusion detection systems (IDS) struggle to maintain high detection accuracy in the presence of high-dimensional data and extreme class imbalance, particularly for rare but critical attacks. The growing proliferation of devices on the Internet of Things (IoT) has made modern networks increasingly vulnerable to sophisticated cyber threats. Traditional intrusion detection systems (IDS) struggle to maintain high detection accuracy in the presence of high-dimensional data and extreme class imbalance, particularly for rare but critical attacks.

1. Introduction

IoT devices have become everywhere, spanning diverse domains such as smart healthcare, smart cities, and intelligent transportation systems. These applications aim to deliver efficient and intelligent services by analyzing large data streams. Consequently, such intensive services impose substantial demands on both computing and communication resources to maintain operational integrity [10]. Security is one of the prime concerns in the Internet of Things, as it enables safe and reliable communication among IoT objects operating in untrusted environments [16]. However, resource-constrained IoT devices are often incapable of processing massive data streams within strict latency bounds. Ensuring security in IoT systems is not limited to protecting confidential information or preventing unauthorized access; attackers may also seek to gain control over devices for malicious purposes. Therefore, the importance of forcing security across all IoT layers, from the physical layer to the application layer, becomes evident [24, 2, 7].

1.1. Security Challenges in Resource-Constrained IoT

In response to these issues, modern literature has concentrated on the development of "adaptive and multi-layered security structures that take into account performance requirements while providing energy efficiency and the capability for mitigating different types of IoT-related security issues." This has paved the way for the development of fog computing and edge computing paradigms that provide proximity computing to IoT devices for low latency while providing security services for these devices. In parallel, the integration of blockchain technology and the implementation of various anomaly detection techniques using machine learning frameworks have garnered significant attention for the purpose of providing trust and real-time anomaly detection for IoT devices. "However, the incorporation of security awareness into system-level decision mechanisms is an intelligent task." Recent literature has emphasized the need for jointly optimizing security compliancy mechanisms for the purpose of providing performance in IoT domains that face issues related to 'static security policies'[23, 10].

Research Objectives

The first intention of the current study is to investigate and understand the limitations faced by the security protocols that are presently being practiced in the IoT system. Over time, the devices part of the IoT are seen to be simple, less expensive, and efficient. Still, most security protocols require a large amount of memory capacity, sufficient support from the battery as well as computational powers. Such protocols are not very suitable for the fundamental principles of the IoT. Based on these limitations of the security protocols presently practiced in the IoT system, the challenges faced by these traditional security protocols are the aim of the study.

*Corresponding author

srinidhi.23mis7130@vitapstudent.ac.in (S. Srinidhi); srinidhi.23mis7130@vitapstudent.ac.in ( ); gopikrishnan.s@vitap.ac.in (Author-3)

ORCID(s): 0009-0001-7322-318X (S. Srinidhi); 0009-0001-7322-318X ( ); 0000-0001-9082-901 (Author-3)

The aim of doing this research is to develop a lightweight and security-aware communication protocol that will be capable of offering various security functionalities such as authentication, confidentiality, authorization, and data integrity to IoT devices in IoT environments. Therefore, since IoT devices possess limited resources in terms of computation, energy, and memory capacity, the aim of achieving this proposed protocol is to adhere to a minimal resource use while still maintaining an acceptable level of security or to achieve a balance between security and resource use while ensuring IoT devices in IoT environments communicate securely without utilizing their scarce resources to perform complex security functionalities.

To establish a security mechanism that can be able to perform more effectively in an IoT environment that does not involve a heavy infrastructure or processing capabilities.Future solutions seek to enhance security further by adding more layers or a centralized component; therefore, it may result in an increase in the likelihood of delaying resource usage.This study seeks to address this issue by introducing a simple, practical, and decentralized security protocol that can be more effective in an IoT environment by fitting into the capabilities of processing, memory, or energy constraints in IoT technology.

Basically, the aim of the study is to verify the performance of the lightweight security-aware protocol through simulation and evaluations as follows.the performance parameters such as costs, efficiency in terms of power, and communication will be conducted and compared.cost and efficiency performance measures such as power and communication will be used to compare the performance of the lightweight security-aware protocol and its effectiveness.simulations and other methods will be used to verify that the processes and methods used indeed result in advances in efficiency and safety on devices used in IoT platforms to extend the lifespan of devices used in the platforms.

2. Literature

2.1. Lightweight Cryptographic and Communication Security

Lightweight cryptographic algorithms are designed to reduce computational and energy overhead while preserving confidentiality and integrity [5]. Modified encryption architectures and efficient key agreement schemes demonstrate feasibility for constrained environments by reducing arithmetic complexity and memory usage [4, 27]. Communication-level optimizations such as lightweight publish–subscribe mechanisms further reduce bandwidth consumption while maintaining reliability [26]. These studies point out the importance of algorithmic simplification to enable practical security deployment in constrained IoT systems.

2.2. Lightweight Intelligent Security Mechanisms for IoT Protocols

Resource-constrained IoT device applications often incorporate lightweight intelligent security solutions as part of security-conscious communication protocols and authentication protocols. These protocols are more intelligent, instead of relying on centralized intrusion detection systems, and involve intelligence at the protocol layer responsible for local anomaly detection, access control, and secure session management [21].Learning models with quantized and pruned learning are used to reduce resource consumption, computation complexity, and energy cost with acceptable security performance. Also, federated learning-based protocols are used to reduce communication complexity through the distribution of updates without exchanging raw data, thus improving privacy preservation. Nevertheless, challenges remain with regard to scalability, synchronization costs, and robustness of these systems against adversarial attacks.[3, 25, 6].

2.3. System-Level and Energy-Aware Security Approaches

System-level solutions combine security considerations with scheduling, energy management, and software integrity verification [18]. Energy-aware adaptive sampling and secure edge transmission frameworks demonstrate how operational efficiency and security can be jointly optimized [9, 20]. Hardware–software co-design techniques enable remote attestation and integrity verification with minimal overhead [22]. These approaches make it clear that strong security depends on designs that balance system constraints with realistic threat models.

3. Comparative Literature Overview

3.1. Research Gaps

Existing literature demonstrates significant progress in lightweight security mechanisms; however, most solutions address isolated aspects such as encryption, intrusion detection, or communication efficiency [17]. Limited work focuses on integrated frameworks that jointly optimize security, performance, and energy under real-world deployment constraints [12]. Additionally, scalability and adaptability across heterogeneous IoT environments remain underexplored, particularly in dynamic and adversarial settings [1].

3.2. Problem Statement

Despite the improvement in lightweight cryptography and intelligent security mechanisms, resource-constrained IoT devices continue to face vulnerabilities due to poor security designs and excessive reliance on single-layer protections [19]. The absence of unified, low-overhead security frameworks capable of adapting to varying threat levels and operational constraints limits practical adoption. There is a critical need for security solutions that maintain robustness while parallelymobserving the computational, communication, and energy costs [18].

3.3. Motivation

The motivation for this study arises from the increasing deployment of IoT systems in safety-critical and privacy-sensitive applications [2]. Ensuring dependable security

Table 1
Literature Survey on Lightweight Security Approaches for Resource-Constrained IoT Devices

Ref.	Method Proposed	Techniques Used	Issues Resolved	Limitations
[13]	Federated learning framework survey for IoT	Federated learning, communication optimization, decentralized training	Privacy preservation and scalable learning in constrained IoT	High communication overhead and system heterogeneity
[4]	Lightweight end-to-end encryption scheme	Modified Feistel structure, symmetric cryptography	Reduced computation and energy consumption with strong encryption	Limited evaluation against advanced cryptanalytic attacks
[5]	LRBC lightweight block cipher	SPN–Feistel hybrid cipher design	Low power, minimal area overhead, strong attack resistance	Limited flexibility for dynamic key management
[25]	Quantized autoencoder-based IDS	Model quantization, anomaly detection	Reduced memory and CPU usage for intrusion detection	Dataset dependency and limited adaptability to novel attacks
[14]	Hybrid lightweight intrusion detection system	CNN-BiLSTM hybrid model, model compression	High detection accuracy with low computational overhead	Model drift and adversarial robustness not fully addressed
[21]	Secure D2D group communication protocol	Secret sharing, Chebyshev polynomials, CRT	Low-overhead secure group communication for IoT	Key management complexity in large-scale networks
[27]	Authenticated key agreement protocol	ECC, anonymous authentication, formal verification	Secure key establishment with low device-side computation	Relies on semi-trusted server assumptions
[15]	Lightweight authentication framework	CRC-based information authentication, ECC	Reduced power, computation, and communication overhead	Limited scalability evaluation for large IoT deployments
[22]	Remote software integrity verification survey	Hardware–software co-design, attestation techniques	Low-overhead integrity verification for constrained devices	Deployment complexity and limited expressiveness
[28]	Waveform-defined communication security framework	Multi-band waveform design, spectral ambiguity	Resistance to AI-based eavesdropping with low power usage	Requires specialized hardware and channel assumptions

under stringent resource constraints is essential to sustain trust and system reliability. By synthesizing insights from lightweight cryptography, machine learning, and system-level optimization, this work aims to contribute toward scalable and efficient security frameworks suitable for next-generation IoT ecosystems [10].

4. Conclusion

This work nails the emphasis on lightweight security-aware protocol design as a basis for computation, memory, and energy-constrained IoT settings. Various research pieces have established that conventional cryptography mechanisms and centralized security mechanisms entail excessive overhead, rendering them inadequate for constrained devices. In turn, lightweight authentication, encryption, and integrity mechanisms, together with edge-assisted intelligence and optimized communication protocols, have emerged as viable alternatives that ensure significant reduction of resource consumption with acceptable security guarantees.[15, 25, 13, 8, 14**?**]. Despite these advances, challenges such as scalability, trust management, and dynamic threat adaptation persist, highlighting the need for integrated and adaptive lightweight security frameworks.[20, 1, 21, 27, 11]

Declarations

Ethical Approval

This manuscript reports studies that do not involve human participants, human data, human tissue, or animals.

Conflict of Interest

The authors have no conflict of interest to declare that are relevant to the content of this article.

Authors' Contributions

S. Gopikrishnan contributed to the conceptualization, Formal analysis, drafting the original manuscript, and designing the experimental protocols. Author-2 was responsible for Conceptualization, methodology, software implementation, and dataset curation. Author3 contributed to conceptualization, supervision, and in-depth analysis of the experimental results. Author-3 handled the review and editing of the final draft, as well as the overall validation of the study results.

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Availability of data and materials

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